Advanced Data Management

Query Answering in (G)LAV
Virtualization

Domenico Fabio Savo

Corso di laurea magistrale
INGEGNERIA INFORMATICA
Dipartimento di Ingegneria Gestionale, dell'Informazione e della Produzione
University of Bergamo

Outline of the course

1. Introduction to propositional logic
2. Introduction to First-order Logic
3. Relational Calculus
4. **I**nformation **I**ntegration **S**ystems (IIS)
5. Logical formalization of IISs
6. Mapping between **Global Schema** e **Data Sources**
7. Incomplete information databases
8. **Query answering over IISs**
9. **O**ntology-**B**ased **D**ata **M**anagement (OBDM)
10. Description Logic Ontologies
11. Query answering in ontologies
12. Query answering in OBDM

Query Answering in (G)LAV

Logical Formalization of Information Integration Systems – Syntax

An information integration system (IIS) is a triple **J** = $\langle G, M, S \rangle$ where:

- **G** is the **global schema**
  - A logical theory over a relational alphabet A_G .
- **S** is the **source schema**
  - A logical theory over a relational alphabet A_S.
- **M** is the **mapping between S and G**
  - A logical theory over the relational alphabet $A_S \cup A_G$.

Logical Formalization of I.I.S.s – Semantics

Assume a information integration system ⟨**J**, **D**⟩, where **J** = ⟨**G**, **M**, **S**⟩ is an information integration specification and **D** is a **source database** for **S**

The semantics of ⟨**J**, **D**⟩ is defined as follows:

sem(**J**,**D**) = { **B** | **B** is a **global database** s.t. (**B**,**D**) ⊨ **M** }

Assume a query q over A_G. The **certain answers** of q w.r.t. **J** and **D** are:

cert(**q**,**J**,**D**) ={ $\bar{a}$ | $\bar{a}$ ∈ **q**(**B**) for each **B** ∈ sem(**J**,**D**)}

Query Answering – Recognition Problem

- **DEF:** **QA-IIS** is the following decision problem:
  - **Input**: an I.I.S. J = $\langle$**G**, **M**, **S**$\rangle$, a FOL query $\boldsymbol{q(\bar{x})}$, a source database $\boldsymbol{D}$, a tuple of constants $\bar{c}$
  - **Question**: Is $\bar{c} \in cert(\boldsymbol{q}, J, \boldsymbol{D})$?

Conjunctive Query (recap)

- Let $\varphi(x_1, \dots, x_n)$ be a FOL formula with free variables $x_1, \dots, x_n$

- $\varphi(x_1, \dots, x_n)$ is an Existential Conjunction if it is of the form

$$\exists y_1, \dots, \exists y_n \, P_1(\bar{z}_1) \wedge \dots \wedge P_k(\bar{z}_k)$$

- A FOL query $\{(x_1, \dots, x_n) | \varphi(x_1, \dots, x_n)\}$ is a **Conjunctive Query** if $\varphi(x_1, \dots, x_n)$ is an Existential Conjunction.

Union of Conjunctive Queries (recap)

- A formula $\varphi(x_1, \dots, x_n)$ is Union of Existential Conjunctions if it is of the form

$$\bigvee_i \psi_i(x_1, \dots, x_n)$$

where each ψ_i is an existential conjunction.
Observe: same free variables.

- A FOL query $\{(x_1, \dots, x_n) | \varphi(x_1, \dots, x_n)\}$ is a **Union of Conjunctive Queries** if $\varphi(x_1, \dots, x_n)$ is a Union of Existential Conjunctions.

Our Assumptions

As made for GAV, we assume:

- That **G** is an empty theory over the alphabet $\boldsymbol{A_G}$, i.e., **G** is a database schema without integrity constraints.
- That **S** is an empty theory over the alphabet $\boldsymbol{A_S}$, i.e., **S** is a database schema without integrity constraints.
- That **GLAV mappings** are defined via queries that we can actually handle: **Conjunctive Queries**

From now on, if not otherwise stated, we always consider the assumptions made above to be true moreover, if not otherwise stated, we will assume that the user queries are conjunctive queries

Local As View (recap)

LAV mappings describe the **source schema** in terms of the global schema

A LAV **mapping assertion** is a pair $\langle \boldsymbol{s}, \boldsymbol{q_g} \rangle$ where:

- $\boldsymbol{s}$ is a **single predicate** name of arity **n** in $\boldsymbol{A_S}$
- $\boldsymbol{q_g} = \{\bar{x} \mid \varphi_g(\bar{x})\}$ is a FOL **query of arity n over** $\boldsymbol{A_G}$

Logical formalization of LAV mapping assertions:

$$\forall \bar{x} \,.\, \boldsymbol{s(\bar{x})} \rightarrow \boldsymbol{\varphi_g(\bar{x})}$$

A LAV mapping is a **finite** set of LAV mapping assertions

Global and Local As View (recap)

Materialization techniques for **LAV** mapping are similar to those for **GLAV**, so we will look at the more general case of **GLAV**

A GLAV **mapping assertion** is a pair $\langle \boldsymbol{q_s}, \boldsymbol{q_g} \rangle$ where:

- $\boldsymbol{q_s} = \{\bar{x} \mid \varphi_s(\bar{x})\}$ is a FOL query of arity **n** over $\boldsymbol{A_S}$
- $\boldsymbol{q_g} = \{\bar{x} \mid \varphi_g(\bar{x})\}$ is a FOL query of arity **n** over $\boldsymbol{A_G}$

Logical formalization of GLAV mapping assertions:

$$\forall \bar{x}.\, \boldsymbol{\varphi_s(\bar{x})} \rightarrow \boldsymbol{\varphi_g(\bar{x})}$$

A GLAV mapping is a **finite** set of GLAV mapping assertions.

Conjunctive GLAV Mapping

A **Conjunctive GLAV mapping assertion** is a pair $\langle \boldsymbol{q_s}, \boldsymbol{q_g} \rangle$ where:

- $\boldsymbol{q_s} = \{\bar{x} \mid \varphi_s(\bar{x})\}$ is a **CQ** of arity **n** over $\boldsymbol{A_S}$
- $\boldsymbol{q_g} = \{\bar{x} \mid \varphi_g(\bar{x})\}$ is a **CQ** of arity **n** over $\boldsymbol{A_G}$

A **Conjunctive GLAV mapping** is a **finite** set of Conjunctive GLAV mapping assertions.

Query Answering – Recognition Problem

- Let **CQ-GLAV** be the class of **conjunctive GLAV mappings**

- **DEF: UCQ-QA-IIS($\mathcal{L}$)** where $\mathcal{L}$ is a class of mappings is the following problem
  - **Input**: An I.I.S. $J = \langle \boldsymbol{G}, \boldsymbol{M}, \boldsymbol{S} \rangle$ where $\boldsymbol{G}$ and $\boldsymbol{S}$ are empty theories and $\boldsymbol{M}$ is a mapping in $\mathcal{L}$, a source database $\boldsymbol{D}$, a UCQ $\boldsymbol{q}$, and a tuple of constants $\bar{c}$
  - **Question**: Is $\bar{c} \in cert(\boldsymbol{q}, J, \boldsymbol{D})$?

- **Thm: UCQ-QA(CQ-GLAV)** is **decidable**
  - Via materialization (*see the slide on GLAV via materialization*)
  - Via virtualization (we will provide a proof via **virtualization**)

Homomorphisms (recap)

The notion of homomorphism

Let **A** and **B** be two database instances over the same schema

A ***homomorphism*** from **A** to **B** is a function h from dom(**A**) to dom(**B**) (dom(A) and dom(B) can contain both *constants* and *variables\labeled nulls*) satisfying the following properties:

1. For every constant c ∈ Const(**A**), h(c) = c
2. For every R($\bar{a}$) ∈ **A**, we have R(h($\bar{a}$)) ∈ **B**

For a tuple $\bar{a} = (a_1, \ldots, a_n)$ of constants and labeled nulls, $h(\bar{a}) = (h(a_1), \ldots, h(a_n))$. In other words, condition 2. always ensures h(**A**) ⊆ **B** whenever h is a **homomorphism** from **A** to **B**, where h(**A**) = {R(h($\bar{a}$)) | R($\bar{a}$) ∈ **A**}

Composition of Homomorphisms

Let h be an homomorphism from A to B, and let h' be an homomorphism from B to C. The **composition** $h' \circ h$ of h, h' is the mapping defined as follows: $h' \circ h(x) = h'(h(x))$

Thm: The composition $h' \circ h$ is itself a homomorphism h'' from A to C.

Proof: Let $\alpha \in A$. Then, $h(\alpha) \in B$ and therefore $h'\big(h(\alpha)\big) \in C$. The claim follows straightforwardly.

Finding Homomorphisms

Finding **homomorphisms** is harder than it may seem

- **Thm:** Given two sets of atoms A and B, checking whether there exists a homomorphism from A to B is NP-Complete
- **Thm:** Let A be a set of atoms. For an input set of atoms B, checking whether there exists a homomorphism from A to B is in LOGSPACE
  - Indeed, if we fix A, the number of possible assignments we need to test becomes small

Query Answering and Homomorphisms (Complete Databases)

We have following well known property:

Let $q = \{\bar{x} \mid \bigvee_i \psi_i(x_1, \ldots, x_n)\}$ be a UCQ and D be a database instance

Thm: a tuple $\bar{a} \in q(D)$ if and only if there exists $i = 1, \ldots, n$ and an **homomorphism** h from ψ_i to D such that $h(\bar{x}) = \bar{a}$

Query Answering In Information Integration

	Materialized	Virtual
GAV	1	2
LAV	3	4

Virtualization in LAV

We want to produce a query over the **sources** that captures the results of a user query over the **global schema**.

Desired properties of the **source query**.

- We want to use it to compute **certain answers**.
- It should fall into a specific query language.

We cannot use simple unfolding techniques with LAV mappings (as for GAV), since LAV does not define the global schema atoms directly

Indeed, computing a rewriting for LAV is a complex task

Perfect Rewritings

- Assume an Information Integration System $\boldsymbol{J} = \langle \boldsymbol{G}, \boldsymbol{M}, \boldsymbol{S} \rangle$.
  - No further assumptions on the three components for now
- Assume a class of queries ***L*** (a language of queries), e.g., CQs, UCQs, FOL, …
- **Definition**: Given a query q over G, a **perfect *L*-rewriting of q w.r.t. J** is a query $\boldsymbol{rew}_{\boldsymbol{L}}^{(\boldsymbol{q},\boldsymbol{J})}$ in the language ***L*** such that the following condition holds for every database D for S

$$\bar{a} \in cert(\boldsymbol{q}, \boldsymbol{J}, \boldsymbol{D}) \ \ if\ and\ only\ if\ \ \bar{a} \in \boldsymbol{rew}_{\boldsymbol{L}}^{(\boldsymbol{q},\boldsymbol{J})}(\boldsymbol{D})$$

Perfect Rewritings – Example

Consider an information integration specification **J** = ⟨**G**, **M**, **S**⟩, where **M** is defined as follows:

$$\mathbf{M} = \{\forall x, y.\, \boldsymbol{TStudents}(x, y) \rightarrow \exists z.\, \boldsymbol{Person}(x) \wedge \boldsymbol{Code}(x, z) \wedge \boldsymbol{Name}(x, y)\}$$

Consider the following query $q(\bar{x})$ over **G**

$$q = \{\ (x) \mid Person(x)\ \}$$

The following query q' over **S** is a perfect CQ-rewriting of q

$$q' = \{\ (x) \mid \exists y.\, TStudents(x, y)\}$$

Exercise

Consider an information integration specification **J** = ⟨**G**, **M**, **S**⟩, where **M** is defined as follows:

$$\mathbf{M} = \{\forall x, y.\, \boldsymbol{TStudents}(x, y) \rightarrow \exists z.\, \boldsymbol{Person}(x) \wedge \boldsymbol{Code}(x, z) \wedge \boldsymbol{Name}(x, y)\}$$

What is the perfect rewriting of the following queries?

- $q_1(x) = \{\ (x) \mid \exists y.\, \boldsymbol{Code}(x, y)\ \}$
- $q_2(x, y) = \{\ (x, y) \mid \boldsymbol{Code}(x, y)\ \}$
- $q_3(x, y) = \{\ (x, y) \mid \boldsymbol{Name}(x, y)\ \}$

Exercise – Solution

Consider an information integration specification **J** = ⟨**G**, **M**, **S**⟩, where **M** is defined as follows:

M $= \{\forall x, y. TStudents(x, y) \rightarrow \exists z. Person(x) \wedge Code(x, z) \wedge Name(x, y)\}$

What is the perfect rewriting of the following queries?

- $q_1(x) = \{\ (x) \mid \exists y. Code(x, y)\ \}$

  $q'_1(x) = \{\ (x) \mid \exists y. TStudents(x, y)\}$

- $q_2(x, y) = \{\ (x, y) \mid Code(x, y)\ \}$

  $q'_2() = \{\ (\quad) \mid \bot\}$ **(The query always false)**

- $q_3(x, y) = \{\ (x, y) \mid Name(x, y)\ \}$

  $q'_3(x, y) = \{\ (x, y) \mid TStudents(x, y)\ \}$

Perfect Rewritings – Problems

When does a perfect rewriting exists for information integration specifications based on **LAV mappings**?

- The definition is not constructive, and it does not guarantee existence

To study perfect rewritings, we need to fix two languages.

1. The language of **global schema queries** we want to rewrite
2. The language of **source schema queries** we want to use for the rewriting

After fixing a language $\boldsymbol{L_G}$ of **global schema queries** and a language $\boldsymbol{L_S}$ of **source schema queries**, we need to face two problems:

1. Is there a perfect $\boldsymbol{L_S}$-rewriting **for every** query in $\boldsymbol{L_G}$?
2. Is such rewriting computable?

Perfect Rewritings in LAV

Fix <u>**UCQ**</u> as **global query language (L_G)** and **source query language (L_S).**

Is there a perfect UCQ-rewriting for every UCQ in the context of LAV information integration systems?

We can show that such rewriting **always exists** and is **computable**, and we can provide and algorithm for its computation.

The algorithm is based on the following intuitions:

- **Sound Rewritings**: Rewritings that capture only part of the certain answers
- **Maximally-Sound Rewritings**: Rewritings that capture as much as possible in a certain class
- **Query Containment**: Checking whether the answers for a query are always contained in the answers for another.

Sound Rewritings in LAV

Assume an information integration specification **J** = ⟨**G**, **M**, **S**⟩, a query $\boldsymbol{q}$, and a class of queries **L**

- **Definition**: A **sound L-rewriting of q w.r.t. M** is a query $\boldsymbol{q'}$ in $\boldsymbol{L}$ such that, for every database $\boldsymbol{D}$, the following condition holds:

$$\boldsymbol{if}\ \bar{a} \in q'(\boldsymbol{D})\ \boldsymbol{then}\ \bar{a} \in cert(q, J, \boldsymbol{D})$$

Intuitively, sound rewritings are an approximation of perfect rewritings

Observe: such approximations come with no guarantees, indeed, the unsatisfiable query $\{(\quad) \mid \bot\}$ is always a **sound rewriting**

Sound Rewritings in LAV – Example

Consider the query $q(\bar{x}) = \{\ x \mid \exists y. A(x,y)\ \}$
and a mapping **M = {** $\boldsymbol{m_1}, \boldsymbol{m_2}$ **}** such that:

- $\boldsymbol{m_1}$: $\forall x. R(x) \rightarrow \exists y,z. A(x,y) \wedge B(y,z)$
- $\boldsymbol{m_2}$: $\forall x. S(x) \rightarrow \exists y,z. A(x,y) \wedge C(y,z)$

The following are sound CQ-rewritings of $q(\bar{x})$

- $\boldsymbol{q_1 = \{\ x \mid R(x)\}}$
- $\boldsymbol{q_2 = \{\ x \mid S(x)\ \}}$
- $\boldsymbol{q_3 = \{\ x \mid R(x) \wedge S(x)\ \}}$

Maximally-Sound Rewritings in LAV

Assume an information integration specification **J** = ⟨**G**, **M**, **S**⟩, a query q, and a class of queries **L**

- **Definition**: a sound L-rewriting q' of q w.r.t. J is a **maximally-sound L-rewriting of q w.r.t. J** if for every database **D** for **S** there exists no sound L-rewriting q'' of q w.r.t. J such that

$$q'(D) \subseteq q''(D)$$

Intuitively, maximally-sound rewritings are one of the **best possible approximations** of certain answers that one can obtained in a language. Of course, they may not be **unique**

Maximally-Sound Rewritings in LAV – Example

Consider the query $q(\bar{x}) = \{\ x \mid \exists y. A(x,y)\ \}$

and a mapping **M = {** $\boldsymbol{m_1, m_2}$ **}** such that:

- $\boldsymbol{m_1}$: $\forall x. R(x) \rightarrow \exists y, z. A(x,y) \land B(y,z)$
- $\boldsymbol{m_2}$: $\forall x. S(x) \rightarrow \exists y, z. A(x,y) \land C(y,z)$

The following are **maximally-sound CQ-rewritings** of $q(\bar{x})$

- $\boldsymbol{q_1 = \{\, x \mid R(x) \}}$
- $\boldsymbol{q_2 = \{\, x \mid S(x)\, \}}$

The following sound CQ-rewriting of $q(\bar{x})$ is **not maximal**.

- $\boldsymbol{q_3 = \{\ x \mid R(x) \land S(x)\ \}}$

Computing Perfect UCQ-Rewritings

Consider an information integration specification **J** = ⟨**G**, **M**, **S**⟩ with **M** a LAV mapping, and a UCQ **q** over **G**.

- **Definition**: Let q_m be the UCQ defined as the **union of all** the maximally-sound CQ rewritings of q w.r.t. **J**

Then we have the following result.

- **Thm**: q_m is a **perfect UCQ-rewriting** of q w.r.t. **J**

Computing Perfect UCQ-Rewritings – Proof

Thm: q_m is a **perfect UCQ-rewriting** of q w.r.t. **J**

Proof: We need to prove that $\bar{a} \in cert(q, J, D)$ if and only if $\bar{a} \in q_m(D)$.
Assume $q_m = \{\bar{x} \mid \bigvee_i \phi_i(\bar{x})\}$.

If part. We need to prove: **If $\bar{a} \in q_m(D)$ then $\bar{a} \in cert(q, J, D)$**

- Suppose $\bar{a} \in q_m(D)$. So, there exists a query $q_i \in q_m$ such that $\bar{a} \in q_i(D)$. By construction, we know that q_i is a sound rewriting of q, which means that $\bar{a} \in cert(q, J, D)$.

Only if part. We need to prove: **If $\bar{a} \in cert(q, J, D)$ then $\bar{a} \in q_m(D)$**

- This part of the proof is more complex. An intuition follows. (full proof in [S. Abitebul & O.M. Duschka. PODS 1998]. See Proof of Theorem 4.2.)
  If $\bar{a} \in cert(q, J, D)$ then there exists a CQ q_i over S such that
  1. $\bar{a} \in q_i(D)$, and
  2. q_i is a **sound CQ-rewriting** of q w.r.t. J.

  Since, by construction, q_i is a sound rewriting, then q_i is a disjunct in q_m. So, $\bar{a} \in q_m(D)$

Checking Sound Rewritings

From the theorem above, in order to build a **perfect rewriting** we can:

1. Compute **all maximally sound rewritings** of the given query
2. Output the **union** of all these rewritings

This is still not an algorithm!

1. **How can we generate the maximally sound rewritings?**
2. **How many of them are there?**

To answer these questions we must first introduce the following notions:

- **Query Expansion**
- **Query Containment**

Expansion

Consider an information integration specification **J** = ⟨**G**, **M**, **S**⟩ wiht **M** a LAV mapping, and a CQ q_S over **S**.

Def: The **expansion** $\mathbf{exp}^{(q_S, M)}$ **of** q_S **w.r.t.** M is the query over **G** obtained by replacing each atom $R(\bar{z})$ in q_S with the view q_R over **G** such that $\langle R, q_R \rangle \in M$

Observe: $\mathbf{exp}^{(q_S, M)}$ is a conjunctive query!

Intuitively, we use M as a **reverse mapping** from the **schema** to the **global schema** and unfold q_S over M in a <u>GAV fashion</u>.

Expansion – Example

Assume the following queries over a source schema **S**.

- $q_1 = \{ x \mid R(x) \}$
- $q_2 = \{ x \mid R(x) \land S(x) \}$

Assume the mapping M containing the following assertions.

- $m_1 : \forall x. R(x) \rightarrow \exists y, z. A(x, y) \land B(y, z)$
- $m_2 : \forall x. S(x) \rightarrow \exists y, z. A(x, y) \land C(y, z)$

The following are the expansions of q_1 and q_2 w.r.t. M.

- $\mathbf{exp}^{(q_1, M)} = \{ x \mid \exists y, z. A(x, y) \land B(y, z) \}$
- $\mathbf{exp}^{(q_2, M)} = \{ x \mid \exists y, z, v, w. A(x, y) \land B(y, z) \land A(x, v) \land C(v, w) \}$

Query Containment

Assume CQs $\boldsymbol{q} = \{\, \overline{\boldsymbol{x}}_{\mathbf{1}} \mid \boldsymbol{\varphi}_{\mathbf{1}}\,(\overline{\boldsymbol{x}}_{\mathbf{1}})\,\}$ and $\mathbf{q}' = \{\overline{\boldsymbol{x}}_{\mathbf{2}} \mid \boldsymbol{\varphi}_{\mathbf{2}}\,(\overline{\boldsymbol{x}}_{\mathbf{2}})\}$ over a schema G

Definition: Let q and q' be two query of the same arity over a schema S. **Query containment** is the problem of checking whether $\mathrm{q}(D) \subseteq q'(D)$, for every database **D** for **S**. We write $\boldsymbol{q} \sqsubseteq \boldsymbol{q}'$ to denote that q is contained in q'.

Thm: Let q e q' be two CQs query of the same arity over **S**. We have that $\boldsymbol{q} \sqsubseteq \boldsymbol{q}'$ if and only of there exists a **homomorphism** h from q' to q such that $h(\bar{x}_1) = \bar{x}_2$

Checking Sound Rewritings

Consider an information integration specification **J** = ⟨**G**, **M**, **S**⟩ wiht **M** a LAV mapping, a CQ q_S over **S** and CQ q_g over the global schema **G**.

We can use expansion to check sound rewritings:

Thm: q_S is a **sound rewriting** of q_G if and only if $\mathbf{exp}^{(q_s,M)} \sqsubseteq q_G$, i.e. $\mathbf{exp}^{(q_s,M)}(D) \subseteq q_G(D)$, for every database D for **G**

Proof: The (complex) proofs uses the fact that $\exp^{(q_s,M)}$ mimics the chase

Size of Maximally-Sound CQ-Rewritings in LAV

Consider an information integration specification **J** = ⟨**G**, **M**, **S**⟩ with **M** a LAV mapping, and a CQ q_G over **G**. The previous **Theorem** provides a way to check if a query q_S over **S** is a **sound CQ-rewriting** of q_G.

What about maximally-sound CQ-rewritings?

We have the following:

- **Thm**: Let $q_G = \{\overline{x} \mid \varphi(\overline{x})\}$ be a CQ over **G** such that φ consists of **n distinct atoms**. If a CQ $q_S = \{\overline{x} \mid \psi(\overline{x})\}$ is a **maximally-sound CQ-rewriting of** q_G w.r.t. **J**, then ψ is logically equivalent to an existential conjunction of **at most n distinct** atoms.

This provides us with an algorithm for computing the set of maximal conjunctive rewritings

The Perfect Rewriting Algorithm

`Input`: An I.I.S. **J** = ⟨**G**, **M**, **S**⟩ where M is a Conjunctive LAV mapping, and a CQ $q_g = \{\bar{x} \mid \varphi(\bar{x})\}$ where φ has **n** conjuncts.

`Output`: A UCQ Q over **S**

$Q \leftarrow \emptyset$

For each CQ $q' = \{\bar{x} \mid \psi(\bar{x})\}$ **over S with at most n conjuncts**

If q' **is a sound rewriting of** q_g**, i.e., if** $exp^{(q',M)} \sqsubseteq q_g$

$Q \leftarrow Q \cup \{q'\}$

End If;

End For;

return Q**;**

Perfect UCQ-Rewriting in LAV

- **Thm**: Let **J** = ⟨**G**, **M**, **S**⟩ be an LAV information integration specification and q_G be a CQ over **G**. For input **J** and q_G, the **Perfect Rewriting algorithm** computes a perfect UCQ-rewriting of q_G w.r.t. **J**.

- **Proof**: The algorithm computes the union of all the maximally-sound CQ-rewritings of the input query.

- **Thm**: The algorithm runs in EXPTIME w.r.t. the size of q_G.

Dealing with GLAV

We can think of GLAV mappings as the composition of a LAV and a GAV mapping. Indeed, each GLAV mapping $\langle \boldsymbol{q_S}, \boldsymbol{q_G} \rangle$ can be rewritten as

$$\langle \boldsymbol{q_S}, \boldsymbol{p} \rangle \cup \langle \boldsymbol{p}, \boldsymbol{q_g} \rangle$$

In logical terms we split the implications into two different assertions:

GLAV: $\forall \bar{x}. \varphi_S(\bar{x}) \rightarrow \varphi_G(\bar{x})$

$\Downarrow$

GAV: $\forall \bar{x}. \varphi_S(\bar{x}) \rightarrow p(\bar{x})$;

LAV: $\forall \bar{x}. p(\bar{x}) \rightarrow \varphi_G(\bar{x})$

Dealing with GLAV

Assume the GLAV mapping $\mathcal{M}$ containing the following assertions.

- m_1 : $\forall x. \exists w. R(x) \land T(x, w) \rightarrow \exists y, z. A(x, y) \land B(y, z)$
- m_2 : $\forall x. \exists w. S(x) \land K(w, x) \rightarrow \exists y, z. A(x, y) \land C(y, z)$

The following LAV ang GAV mappings **together** are equivalent to $\mathcal{M}$

- **GAV**
  - m_1^G : $\forall x. \exists w. R(x) \land T(x, w) \rightarrow G_1(x)$
  - m_2^G : $\forall x. \exists w. S(x) \land K(w, x) \rightarrow G_2(x)$
- **LAV**
  - m_1^L : $\forall x. G_1(x) \rightarrow \exists y, z. A(x, y) \land B(y, z)$
  - m_2^L : $\forall x. G_2(x) \rightarrow \exists y, z. A(x, y) \land C(y, z)$

Perfect UCQ-Rewriting in GLAV

To deal with GLAV in a virtualized way, we can proceed in three steps.

Consider an information integration specification **J** = ⟨**G**, **M**, **S**⟩ with **M** a **GLAV** mapping, and a CQ q_G over **G**

1. From M, we define an equivalent pair of mappings M_1 and M_2 where M_1 is **LAV** and M_2 is **GAV**.
2. We rewrite q w.r.t. M_1 using the perfect rewriting algorithm for LAV defined above. Let q' be the result of this step.
3. We unforld q' w.r.t. M_2 to obtain a query q'' over the schema **S** (as in GAV)

- **Thm**: q'' is a **perfect UCQ-rewriting** of q w.r.t. **J**.

Query Answering – Recognition Problem

Let **CQ-GLAV** be the class of **conjunctive GLAV mappings**

DEF: UCQ-QA-IIS($\mathcal{L}$) where $\mathcal{L}$ is a class of mappings is the following problem

- **Input**: An I.I.S. $J = \langle \boldsymbol{G}, \boldsymbol{M}, \boldsymbol{S} \rangle$ where $\boldsymbol{G}$ and $\boldsymbol{S}$ are empty theories and $\boldsymbol{M}$ is a mapping in $\mathcal{L}$, a source database $\boldsymbol{D}$, a UCQ q, and a tuple of constants $\bar{c}$
- **Question**: Is $\bar{c} \in cert(q, J, \boldsymbol{D})$?

Thm: UCQ-QA(CQ-GLAV) is **decidable** (actually LOGSPACE in data complexity)

Proof: For a GLAV mapping **M** a query **q** over **G** and database **D** for **S**, we can compute the **perfect UCQ-rewriting q'** of **q** w.r.t. **J** (independently from the size of **D**) and then we can evaluate **q'** over **D** in polynomial time w.r.t. the size of **D** (actually in LOGSPACE).

Credits

The presentations of this module of the course are based on the original slides of Prof. Giuseppe De Giacomo, Prof. Diego Calvanese, Prof. Marco Console, Prof. Maurizio Lenzerini, and Prof. Domenico Lembo.